

Object Oriented Programming

Dr. Imran Ashraf

Office: 02

`imran.ashraf[at]hitecuni[dot]edu[dot]pk`

Are computers dumb?

Review

- Computer?
- Computer programming?

Review (contd.)

- Algorithm?
- Computer Language?

Review (contd.)

- What are three basic ingredients of a programming language?

Review (contd.)

- Constant, variable
- Data types
- If-else, loops
- Functions
- Structures
- Pointers

Reading for Next Week

- Functional Programming
- Structured Programming
- Imperative Programming
- Object Oriented Programming

Structural Programming

- Improved clarity of programs by introducing structured control flow constructs (if/for etc), subroutines, blocks
- Contrast it with ***goto*** programming
- Examples: C, C++, java, python
- Counter Examples: JOSS, FOCAL, assembly languages, MS-DOS batch files, and early versions of BASIC, COBOL, and MUMPS.

Imperative Programming

- An imperative language uses a sequence of statements to determine how to reach a certain goal.
- These statements are said to change the state of the program as each one is executed in turn.
- Examples: C, C++, java, python

C/C++: factorial

- `int factorial = 1`
- `for(int i = 1; i <=n; ++i)`
- `{`
- `factorial *= i;`
- `}`

Functional Programming

- style that treats computation as the evaluation of mathematical functions and avoids changing-state and mutable data
- No local/global state so no side effects
- Makes it easier to understand program
- Examples: Haskell, Scheme, OCaml

Haskell: factorial example

- `factorial n = if n < 2 then 1 else n * factorial (n-1)`

Object Oriented Programming

- OOP is based on the concept of
 - **objects**, which may contain data, in the form of fields, often known as attributes
 - **code**, in the form of procedures, often known as methods.
- Enables:
 - Data encapsulation(hiding)
 - Composition
 - Inheritance
 - Polymorphism

Next week

- Functions
- Passing arguments to functions